These notes follow: "Lectures on the Bethe Ansatz", and is produced by Esha Sury and Dhruv Upreti.

Bethe Ansatz Theory

The Bethe Ansatz, is a collection of methods used to obtain exact solutions to certain quantum many-body systems, especially one-dimensional quantum integrable models. In favorable cases, the Bethe Ansatz gives the exact energy spectrum and corresponding eigenstates by reducing the many-body eigenvalue problem to a set of algebraic equations called the **Bethe equations**.

Intuitively, in models solvable by the Bethe Ansatz, many body scattering is **non-diffractive** meaning that particles retain their individual momenta after collisions, and the full many body scattering matrix factorizes into an ordered product of two body scattering matrices. The Yang Baxter equation is the consistency condition ensuring that the different possible orders in which the same many body scattering process can be decomposed into two body scatterings give equivalent results.

Asymptotic Bethe Ansatz Equations in 1+1 Dimensional Quantum Field Theory

This section discuss the Bethe equations for the spectrum of a generic 1+1 dimensional integrable theory. We consider a theory on a circle, line segment of length L with periodic boundary conditions. In an integrable theory the scattering has features highlighted above:

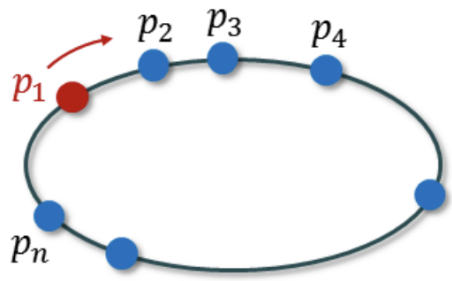
- The number of particles is conserved in any scattering event.
- The momenta do not change but can only be redistributed between particles.
- The S-matrix for multiparticle scattering factorizes; this means the S-matrix for any number of particles is a product of the $2 \to 2$ S matrices.

Starting with a toy model, a theory with only one particle in its particle spectrum of the QFT, such as the sinh-Gordon model for some values of the parameters.

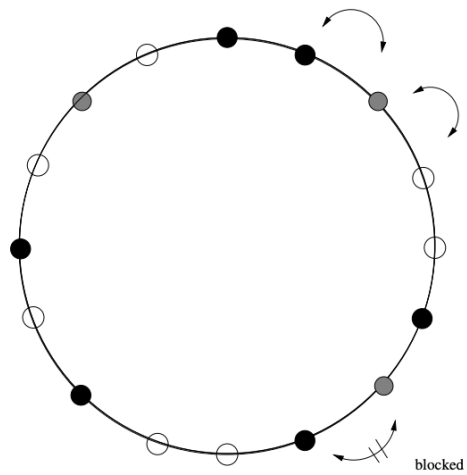
Recall from quantum mechanics, momentum is the generator of spatial translations since $\hat{p} = -id/dx$ (for $\hbar = 1$), then the eigenvalue equation becomes $\hat{p}\,\psi(x) = p\,\psi(x) \Rightarrow \psi(x) = Ce^{ipx}$. Therefore a particle with definite momentum p should have the property that if you translate it by a distance a, it only changes by a phase,

$$\psi(x+a) = e^{ipa}\psi(x).$$

Imagine the first particle is taken around the circle once, eventually bringing it back to its place again. If there w'ere no other particles (or if the theory was non-interacting) the wavefunction would acquire a phase factor e^{ip_1L}, and by the periodicity of the system $e^{ip_1L} = 1$ so $p_1 = 2\pi k/L$ for $k \in \mathbb{Z}$.



(a) Deriving the periodicity condition on a circle. We take the first particle with momentum p_1 around the circle, scattering it through all the other particles.



(b)

Figure 1

Taking into account the interaction with other particles, if L is large compared to the interaction range between the particles the interaction is described by the asymptotic S-matrix (which is just the S-matrix defined using particles that are infinitely far apart before and after scattering since each collision looks approximately like an ordinary infinite line scattering event), for a one particle species integrable theory, the two particle asymptotic S-matrix is just a phase,

$$S(p_1, p_2) = e^{i\alpha(p_1,p_2)}.$$

This is due to the fact that in this toy model with only one particle type in the spectrum and the theory is integrable the number of particles is conserved and the momenta cannot change except possibly being exchanged, so after scattering it only gains a complex phase,

$$|p_1, p_2\rangle_{\text{out}} = S(p_1, p_2)\ |p_1, p_2\rangle_{\text{in}}\,, \quad |S(p_1, p_2)| = 1 \text{ so } S(p_1, p_2) = e^{i\alpha(p_1,p_2)}.$$

When we take the particle around the circle it will scatter through all other particles as shown in Figure 1(a) so we see that,

$$e^{ip_1 L} S(p_1, p_2) S(p_1, p_3) \cdots S(p_1, p_n) = 1$$

For any particle we get,

$$e^{ip_j L} \prod_{k=1, k\neq j}^{n} S(p_j, p_k) = 1, \quad \text{asymptotic Bethe Ansatz equations}$$

This is a set of n algebraic equations for n variables $p_1 \cdots p_n$ which allows us to fix the values of the momenta. In regions where the particles are well separated they propagate freely, so the energy of the state should always be equal to simply the sum of individual particles' energies given as,

$$E = \sum_i \varepsilon(p_i).$$

Note that the asymptotic Bethe equations only describes the energy in a large volume L, more precisely the Bethe equations capture all terms in the large L expansion of the energy which scale as $\sim 1/L^k$ known as the powerlike corrections but miss exponential corrections of the kind $\sim e^{-mL}$ (m is the particle's mass) which physically correspond to the effects of virtual particles propagating around the circle.

In a more general theory one could have a a larger spectrum of particles, and different types could transform into each other during scattering, giving the S-matrix actual matrix structure. Consider a model with K particle types, where each particle can be in a linear combination of different types (or flavors) so its flavor state is in $\mathbb{C}^K$. For n particles their state denoted as A is an element of the tensor product,

$$A \in H_1 \otimes \cdots \otimes H_n, \quad H_i \subset \mathbb{C}^K.$$

Choosing a basis e_i in $\mathbb{C}^K$ we can write,

$$A = \sum_{j_1=1}^{K} \cdots \sum_{j_n=1}^{K} A_{j_1 \cdots j_n} e_{j_1} \otimes \cdots \otimes e_{j_n}.$$

With this we define the S-matrix as a linear operator acting on the tensor product of the two spaces H_i,

$$\hat{S_{ij}} \in \mathrm{End}(H_i \otimes H_j).$$

The S matrix here has four indices, two for incoming particles and two for outgoing particles. If particles one has internal state $e_a \in H_1$ and particles two has internal state $e_b \in H_2$ then before scattering the two particle internal state is $e_a \otimes e_b$ so after scattering,

$$\hat{S}_{12}(p_1, p_2)(e_1 \otimes e_2) = \sum_{c,d} S_{ab}^{cd} e_c \otimes e_d.$$

The Bethe equations for a theory of several particle types is similar to the equation above but the product of the S-matrices now acts on a state $A \in H_1 \otimes \cdots \otimes H_n$, where it's given as,

$$e^{ip_k L} \hat{S}_{k,k+1} \hat{S}_{k,k+2} \cdots \hat{S}_{k,n} \hat{S}_{k,1} \cdots \hat{S}_{k,k-1} A = A, \quad \text{for } k = 1, \cdots, n.$$

Example: 1D Spin-1/2 Heisenberg Chain

This section follows: "On the Theory of Metals. I. Eigenvalues and Eigenfunctions of the Linear Atomic Chain", following Bethe's original formulation of the ansatz for a 1D metal consisting of a linear chain of atoms, each of which has a single s electron with spin.